



KSI
KUMARAGURU
SCHOOL OF
INNOVATION

24ADC001- BUSINESS INTELLIGENCE

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2026-2027
LAB RECORD

**DEPARTMENT OF ARTIFICIAL
INTELLIGENCE AND DATA SCIENCE**

INDEX

Expt. No	Title of the Experiment	Page No	Total Marks	Faculty Signature
1.	Explore the interface and basic features of the BI tool			
2.	Load and visualize a sample dataset.			
3.	Import a dataset into the BI tool. And Cleanse data by handling missing values, outliers, and inconsistencies.			
4.	Transform data to suit BI reporting requirements and Design a dashboard with key performance indicators (KPIs).			
5.	Develop interactive dashboards for dynamic data exploration.			
6.	Integrate data from various sources for comprehensive analysis			
7.	Implement advanced chart types (treemaps, heatmaps, etc.).			
8.	Apply BI tools for forecasting and predictive analytics.			

Average marks

Faculty Signature

EXPT NO:1	EXPLORE THE INTERFACE AND BASIC FEATURES OF BI Tool
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is the purpose of Business Intelligence (BI) tools like Tableau and Power BI?
Business Intelligence tools help collect, analyze, and visualize data. They convert raw data into meaningful insights. They support better business decision-making. They also create interactive reports and dashboards.
2. List any three common data sources that Tableau and Power BI can connect to.
Tableau and Power BI can connect to Excel files. They also support CSV files and SQL databases. Other sources include cloud services and web data. This allows data integration from multiple platforms.
3. What is a dashboard in BI, and how is it different from a single chart?
A dashboard is a collection of multiple visualizations in one place. It provides a complete view of business performance. A single chart shows only one aspect of the data. Dashboards are interactive and support better analysis.
4. Define the following terms:
 - (a). Define Visualization.
Visualization is the graphical representation of data. It uses charts, graphs, maps, and tables. It makes complex data easy to understand. It helps identify patterns and trends quickly.
 - (b). Define KPI.
A KPI (Key Performance Indicator) measures business performance. It helps evaluate progress toward objectives. KPIs are used to monitor important metrics. They support informed decision-making.
 - (c). Define Data Filter.
A data filter displays only selected data based on conditions. It removes unnecessary information from the view. Filters make analysis more focused and efficient. They improve data exploration.
5. Why is it important to clean and transform data before visualization?
Data cleaning removes errors and missing values. Data transformation organizes information into the required format. This improves data quality and accuracy. Reliable data leads to better business insights.
6. What kind of chart would you use to analyze stock price trends over time?
A line chart is best for showing stock price trends. It displays changes over a period of time. It helps identify increases and decreases in prices. It is commonly used in financial analysis.
7. Explain the role of slicers in Power BI and filters in Tableau.

Slicers and filters allow users to view selected data. They help analyze information based on specific conditions. Users can interact with reports easily. This improves decision-making and dashboard usability.

8. What does a calculated field or measure do in BI tools?

A calculated field performs custom calculations on data. It creates new values using formulas. Measures summarize data for analysis. They provide meaningful business insights.

9. What are the advantages of using interactive dashboards for financial data analysis?

Interactive dashboards provide real-time financial insights. They allow users to filter and explore data easily. They improve trend analysis and performance tracking. They support faster business decisions.

10. Which function would you use to find the absolute difference between predicted and actual stock prices?

The ABS() function is used to calculate the absolute difference. It returns only positive values. It measures prediction error accurately. It is useful in financial analysis.

IN-LAB EXERCISE:

Objective:

To understand the basic interface, data import process, and fundamental visualization capabilities of **Tableau** and **Power BI** using a stock prediction dataset, and to explore their usefulness in analyzing financial data.

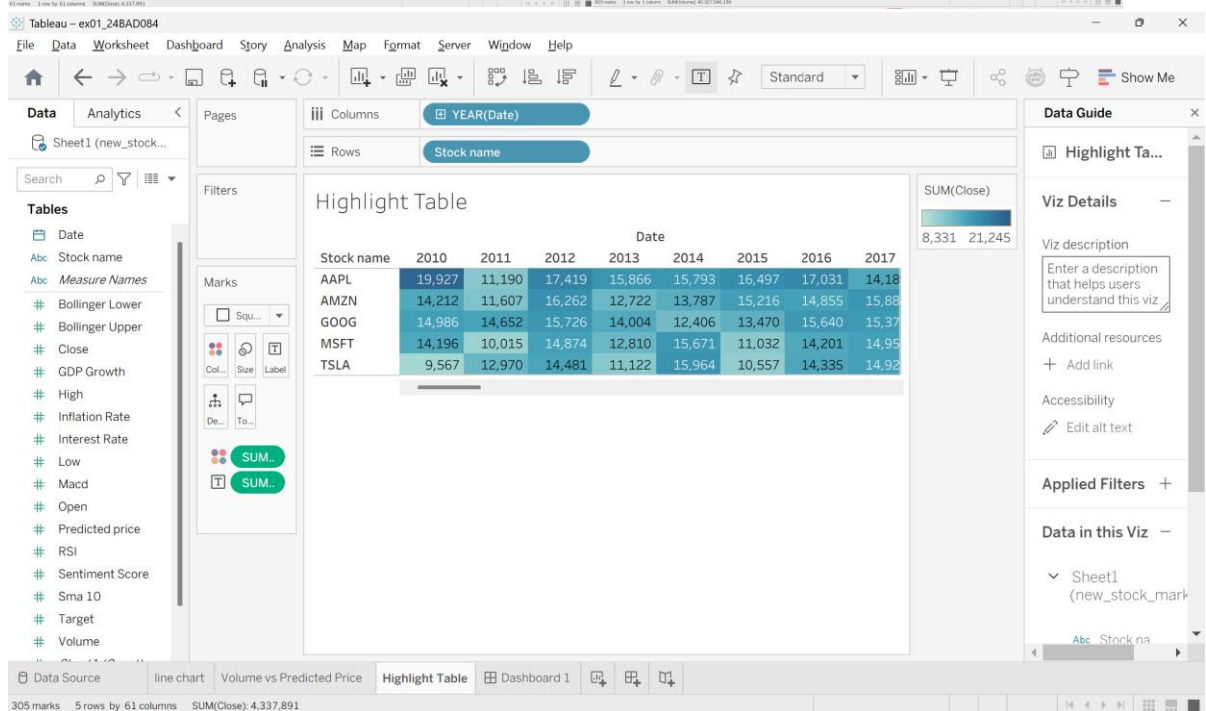
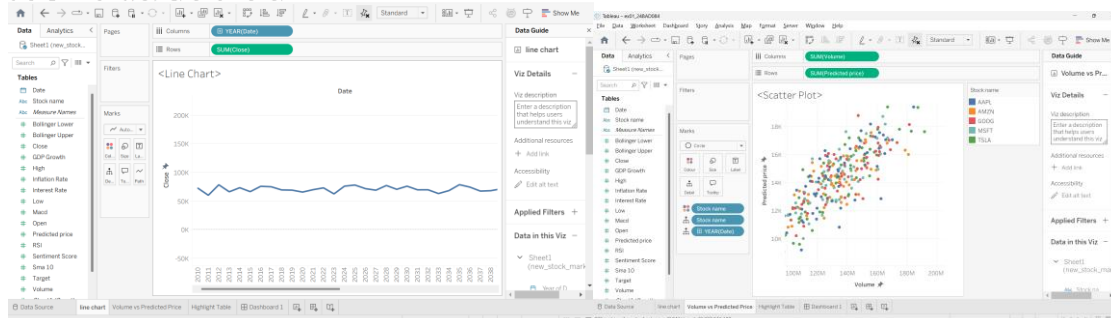
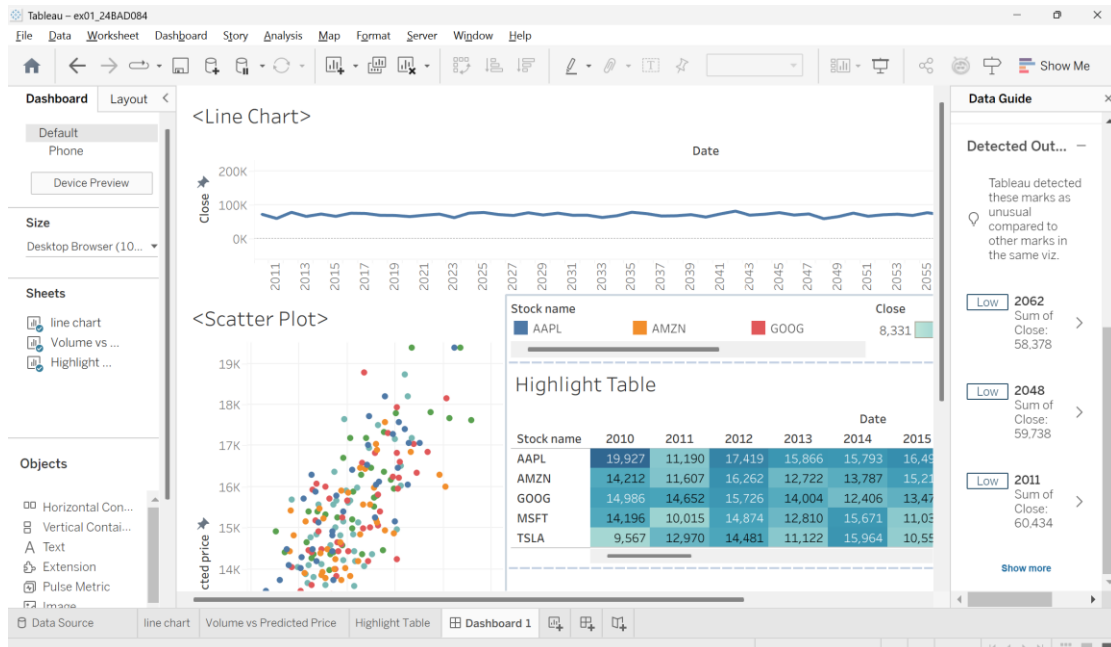
Tools Required:

- Tableau Public / Tableau Desktop
- Power BI Desktop
- Sample Dataset (CSV/Excel) containing:
Date, Stock Name, Open, Close, High, Low, Volume, Predicted Price

Exercise Steps:

Part 1: Working with Tableau

1. **Launch Tableau** and connect to the stock prediction dataset.
2. **Explore the Interface:** ○ Understand components like **Data Pane**, **Sheets**, **Marks Card**, and **Dashboards**.
3. **Create Basic Visualizations:**
 - **Line Chart:** Show Close Price over time. ○ **Scatter Plot:** Plot Volume vs. Predicted Price.
 - **Highlight Table:** Display Predicted Price by stock.
4. **Apply Filters:**
 - Add filters for Stock Name and Date Range.
5. **Create Calculated Field:** ○ Calculate prediction error as $|\text{Predicted} - \text{Close}|$.
6. **Design a Basic Dashboard:**
 - Combine your charts into one interactive dashboard.

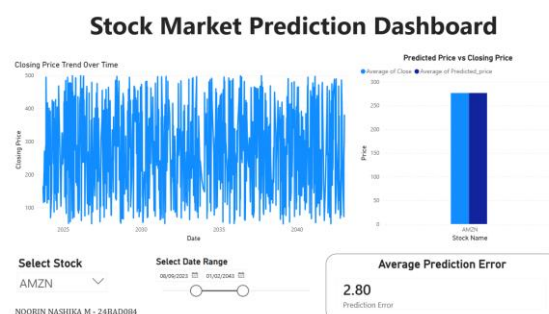
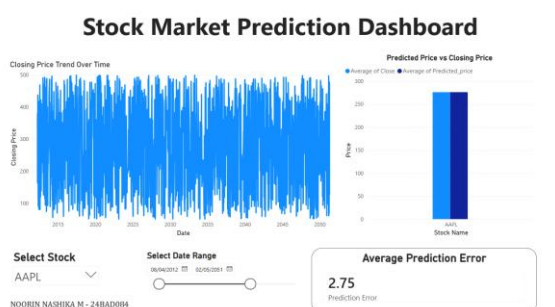
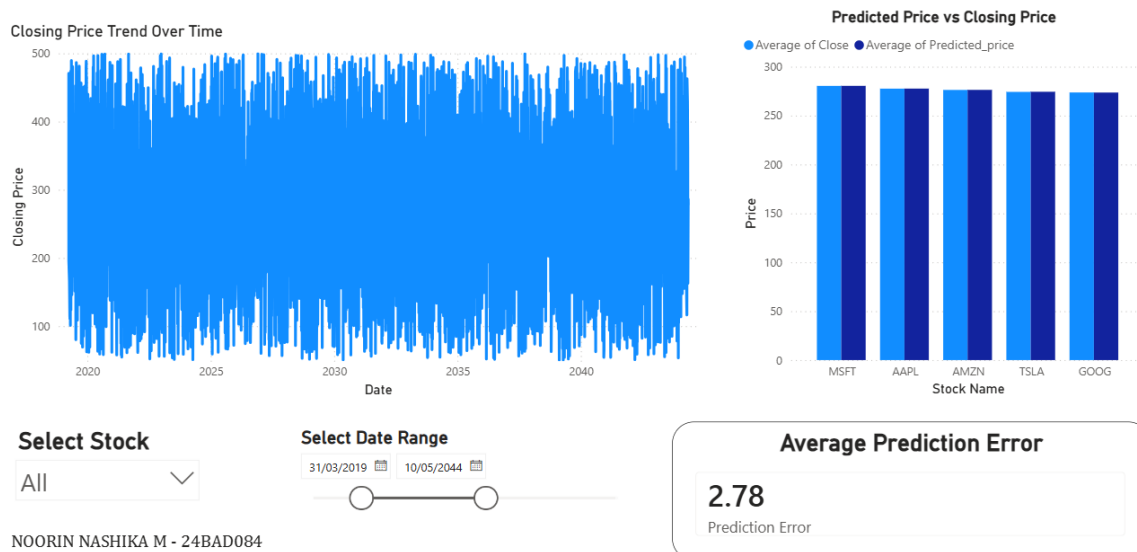


Part 2: Working with Power BI

1. Open Power BI Desktop and import the same dataset.

2. **Explore the Interface:**
 - o Familiarize with **Fields Pane**, **Visualizations Pane**, **Report View**, **Data View**.
3. **Create Basic Visualizations:**
 - o **Line Chart:** Trend of Close Price over time.
 - o **Bar Chart:** Compare Predicted Price vs. Close Price.
 - o **Card (KPI):** Display average prediction error.
4. **Use Filters and Slicers:**
 - o Filter data based on Stock Name and Date.
5. **Create a Simple Measure** using DAX:
 - o Example: Prediction Error = ABS(Predicted - Close)
6. **Build a Report Page:**
 - o Arrange visuals in a single interactive page.

Stock Market Prediction Dashboard



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. Describe your experience with the Tableau interface. What components did you interact with?
The Tableau interface was simple and easy to use. I worked with the Data Pane, Sheets, Marks Card, and Dashboard. These components helped create visualizations. They made data analysis more interactive.
2. How did you import the dataset into Tableau and Power BI?
I selected the Connect option in both tools. Then I chose the CSV/Excel dataset. After loading the file, I verified the data fields. The dataset was ready for visualization.
3. What types of visualizations did you create in both tools? List at least two per tool.
In Tableau, I created a line chart and a scatter plot. In Power BI, I created a line chart and a bar chart. These visualizations showed trends and comparisons. They helped analyze stock data effectively.
4. How did you calculate the prediction error, and how was it displayed in your dashboard?
I calculated the prediction error using $ABS(\text{Predicted Price} - \text{Close Price})$. This formula measured the difference between predicted and actual values. The result was displayed as a KPI card and charts. It helped evaluate prediction accuracy.
5. Which filtering or slicing techniques did you apply, and what insights did they reveal?
I applied filters for Stock Name and Date. This allowed me to analyze selected stocks over specific periods. It revealed changes in stock performance. The filters improved data exploration.
6. Compare the process of creating a line chart in Tableau vs. Power BI. What differences did you notice?
Tableau uses drag-and-drop to create charts. Power BI uses the Visualizations pane. Both tools make chart creation simple and interactive. The interfaces are different but easy to learn.
7. How did you organize the dashboard in Tableau and Power BI? What key elements did you include?
I arranged charts, KPI cards, and filters in a clear layout. Related visualizations were placed together. The dashboard was designed for easy understanding. It provided an interactive view of stock performance.
8. Which tool (Tableau or Power BI) did you find more user-friendly for financial data visualization, and why?
I found Power BI more user-friendly. It has a simple interface and built-in DAX functions. Dashboard creation is quick and efficient. It is suitable for financial data analysis.
9. What challenges did you face while creating visualizations or dashboards?

I faced minor issues while selecting chart types. Creating calculated fields required some practice. Formatting dashboards also took time. These challenges were resolved through experimentation.

10. How can the knowledge gained from this lab be applied to real-world financial data analysis?

The lab helped me understand financial data visualization. It improved my skills in creating dashboards and reports. These skills are useful for analyzing stock market trends. They support data-driven business decisions.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		